How can biomass data inform C model calibration and testing

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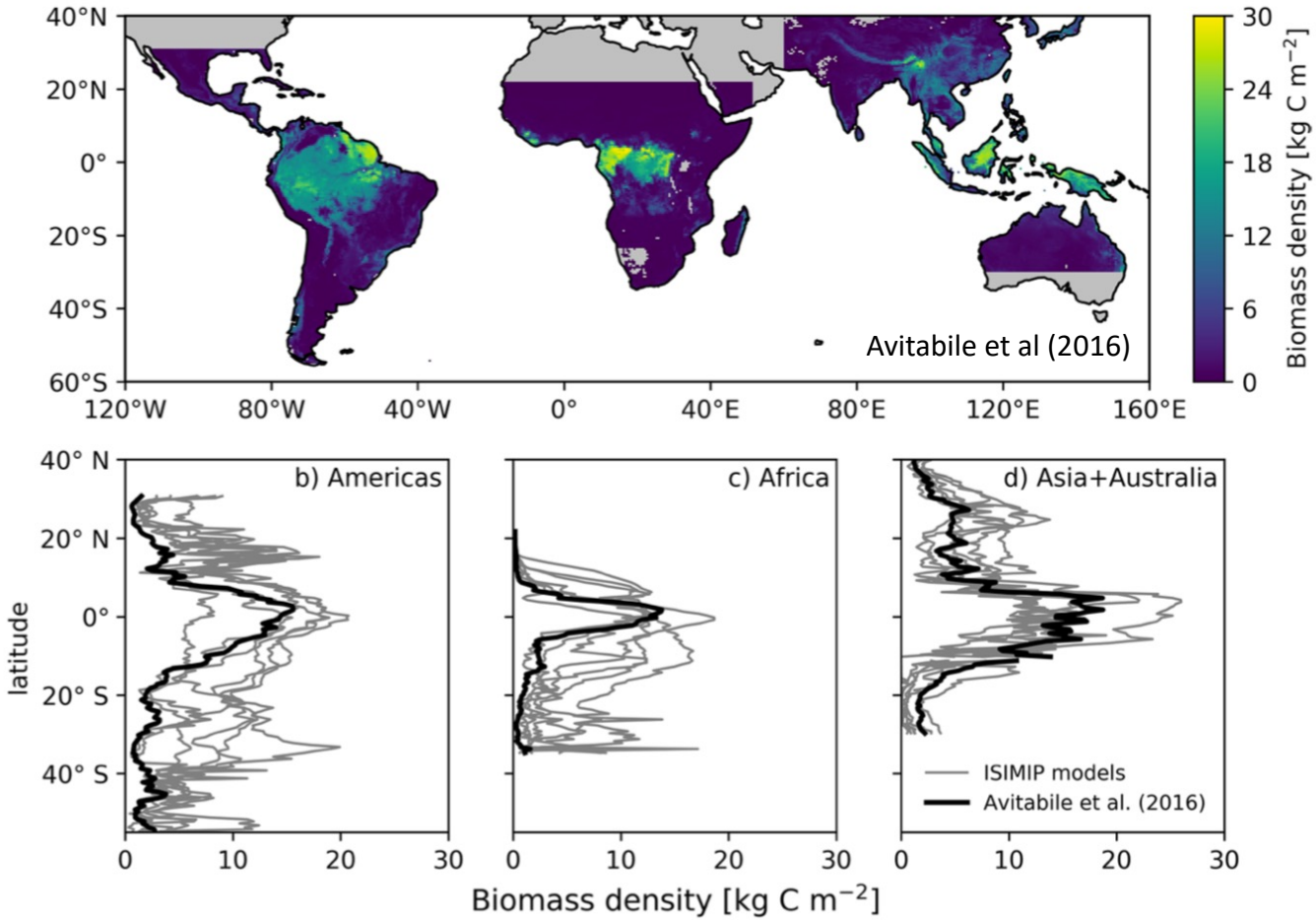
Summary

- Biomass maps can improve
  - Initial conditions for wood C
  - Disturbance histories
  - Steady state turnover estimates of wood C
- Biomass time series can improve
  - Fluxes
  - Woody growth rates
- Challenges

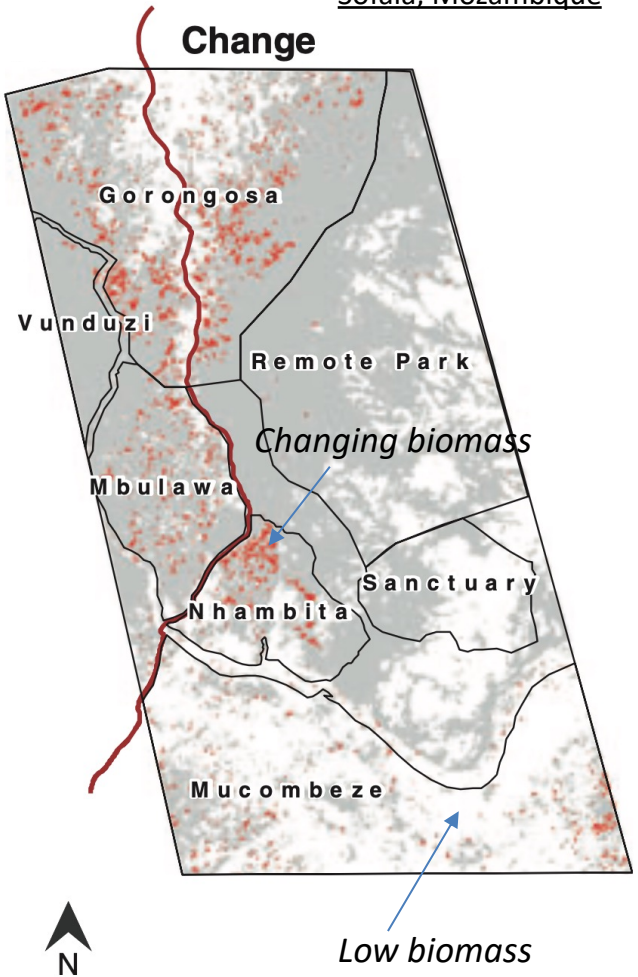
Setting initial conditions

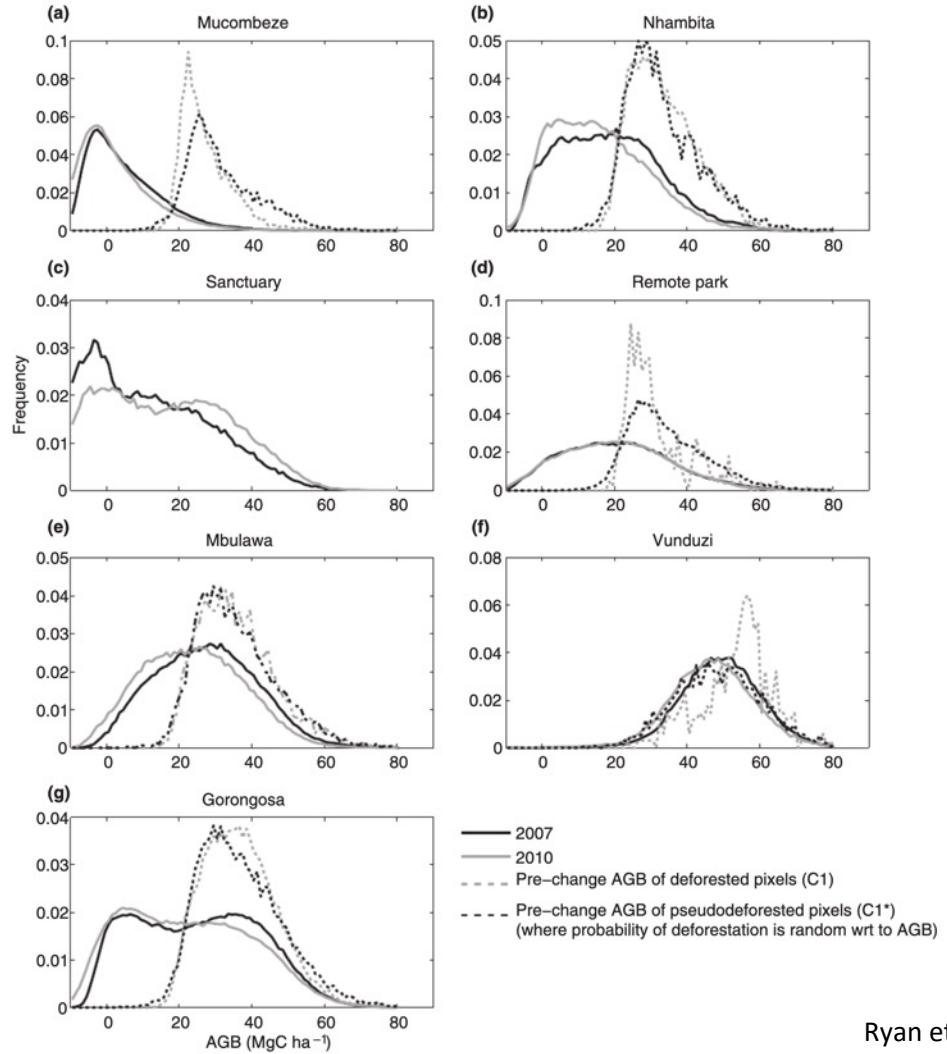


Exbrayat et al. 2019

Disturbance history information

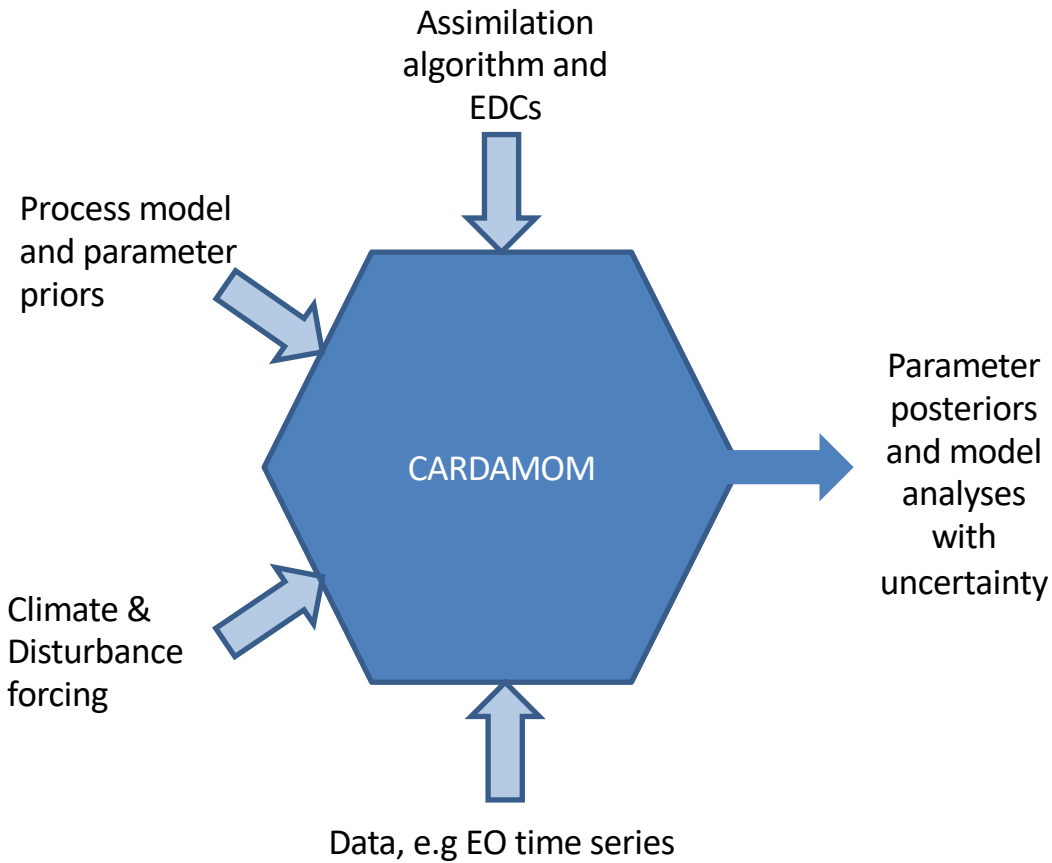
Sofala, Mozambique



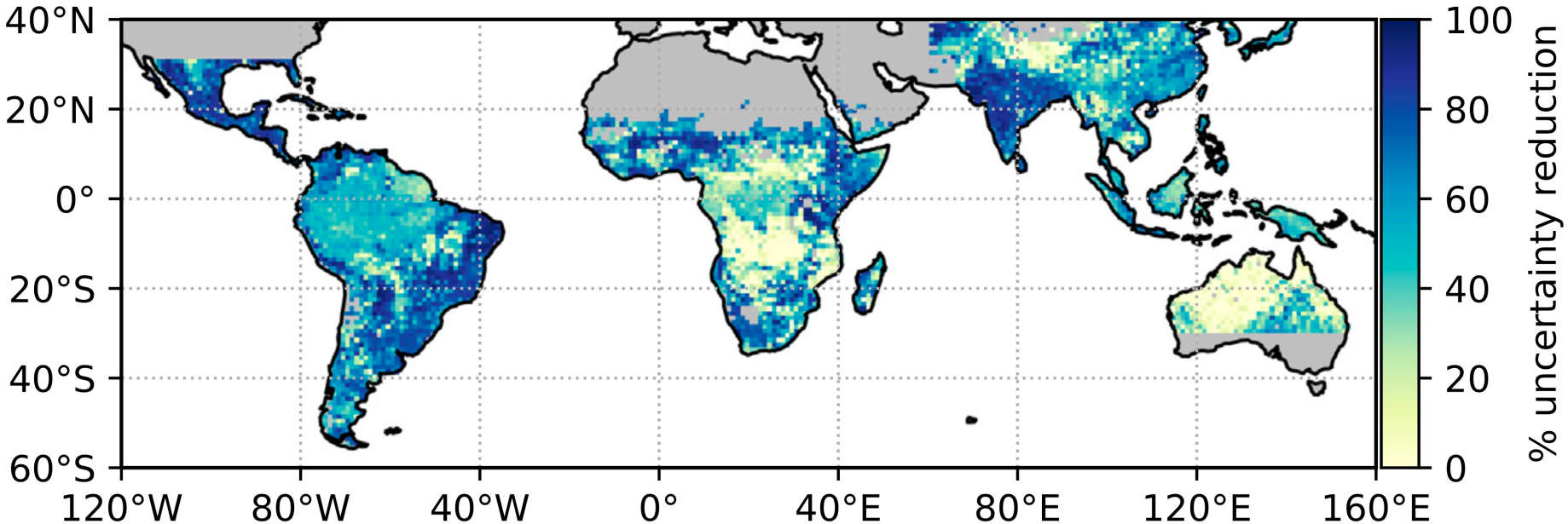


Ryan et al 2012

Model-data fusion Approach



Improved estimates of vegetation C turnover time

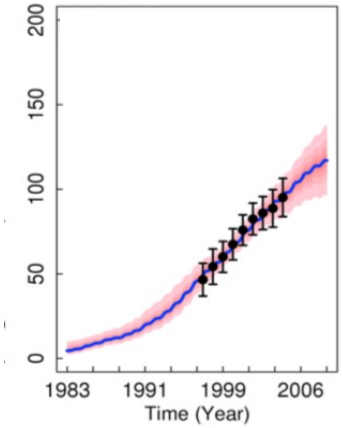


Relative reduction in error of estimated vegetation carbon turnover times when using a prior value for biomass at each pixel compared to a run without

Quegan et al, 2019

Constraining fluxes from biomass information

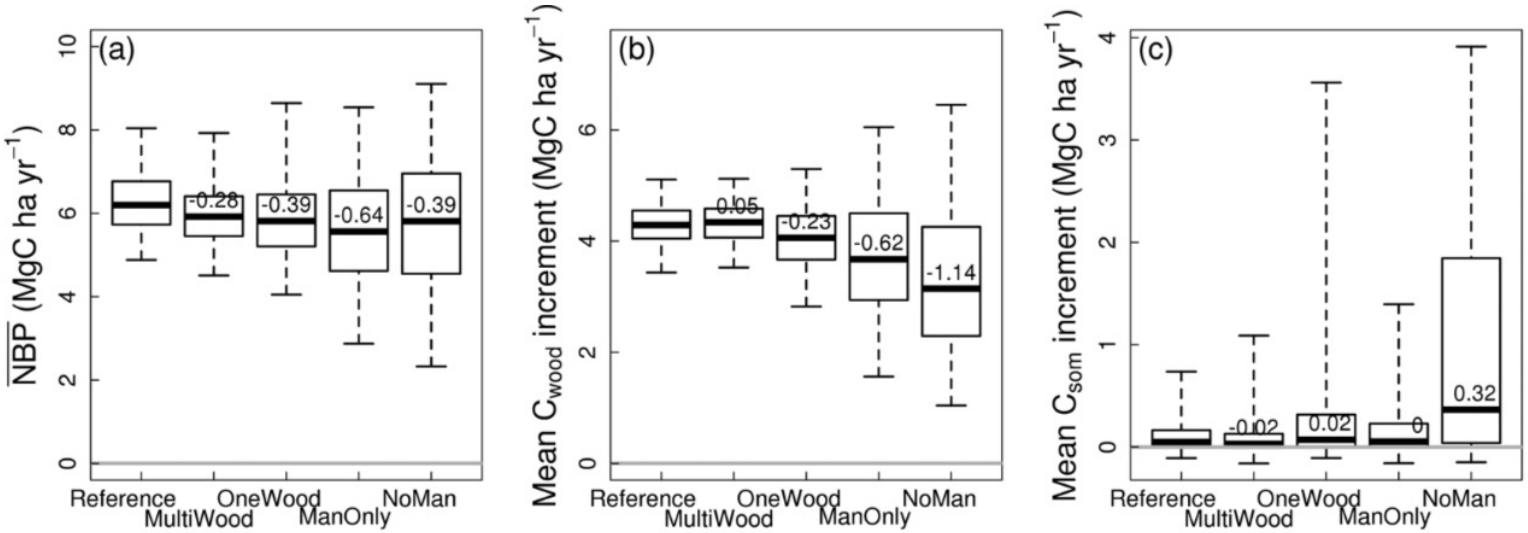
- Aggrading forest, non steady-state
- Duke Forest, NC (Loblolly Pine planted in 1983)
- What information is required to constrain model parameters?



Information content of observations

Data	**Reference**	**Multi Wood**	**One Wood**	**Man Only**	**No Man**
Foliar C/LAI	In situ	MODIS	MODIS	MODIS	MODIS
Root C	In situ	-	-	-	-
Wood C	In situ	annual	one	age	-
Soil C	In situ	HWSD	DWSD	HWSD	HWSD
Concept	*Field site*	*BIOMASS*	*Avitabile*	*LC history*	*Baseline*

Biomass information can constrain NBP, soil dynamics



NBP

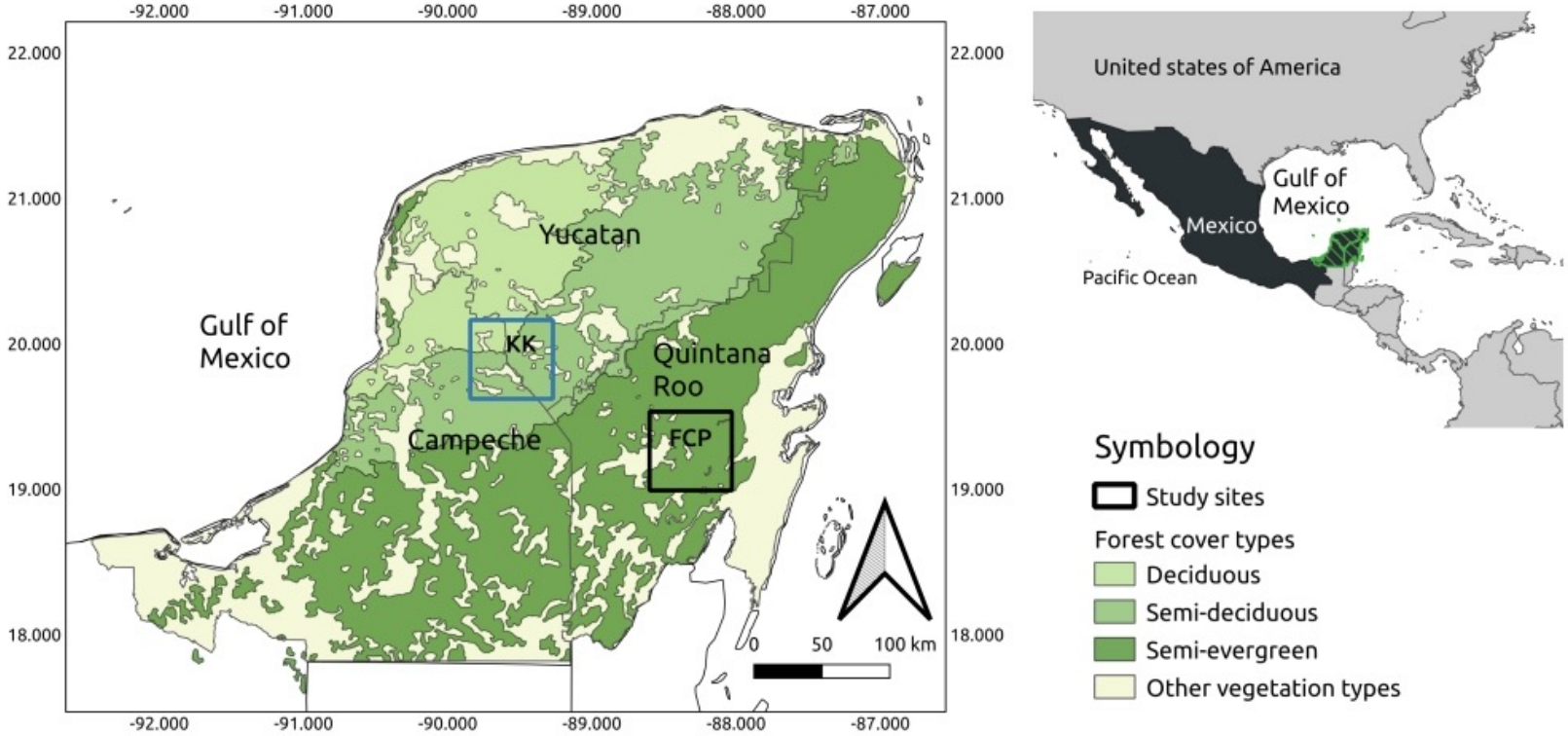
Wood increment

SOM increment

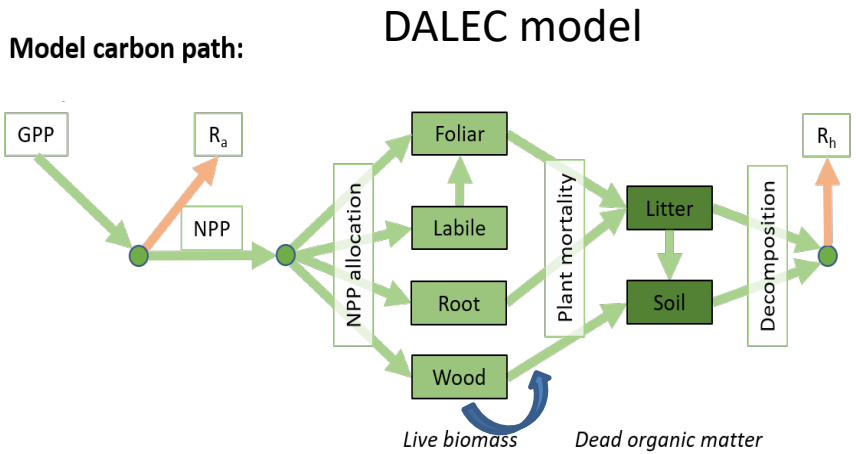
Less assimilated information

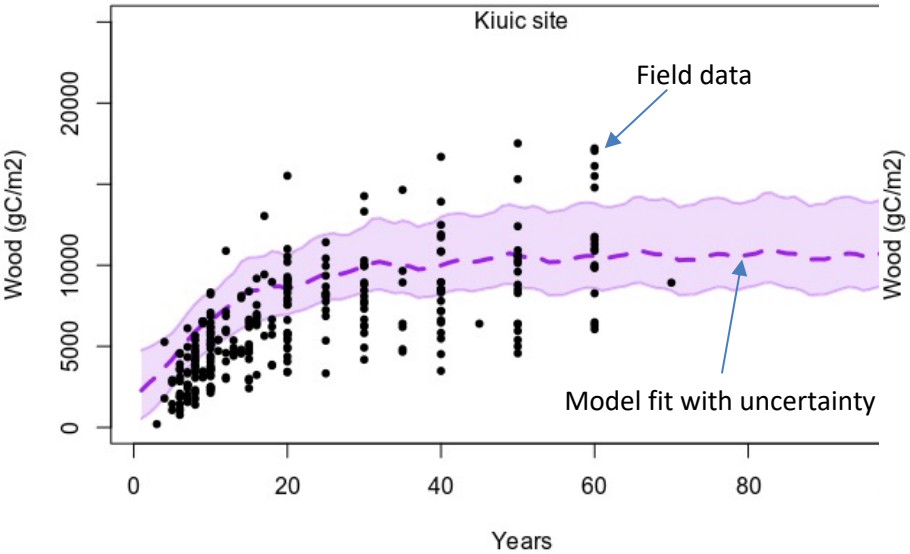
Smallman et al., (2017)

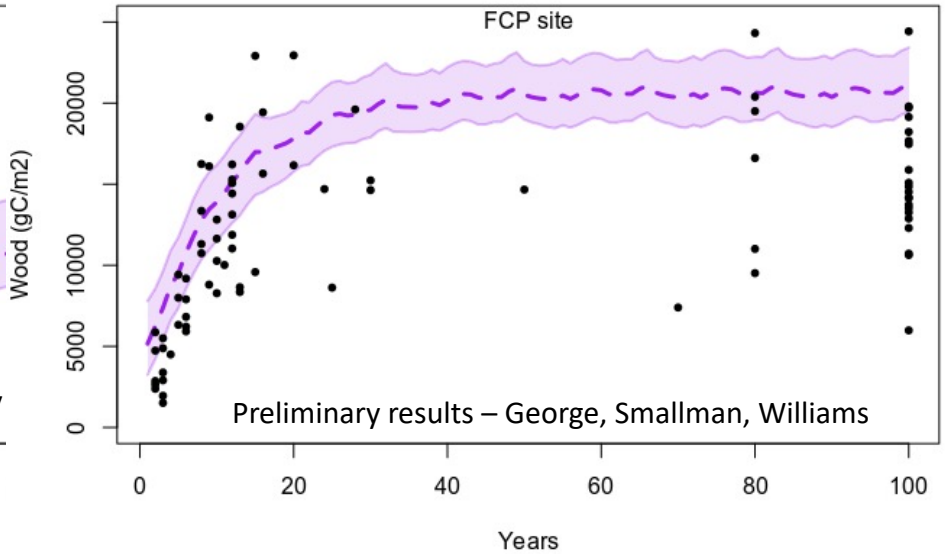
Chronosequence data provide insights into decadal dynamics



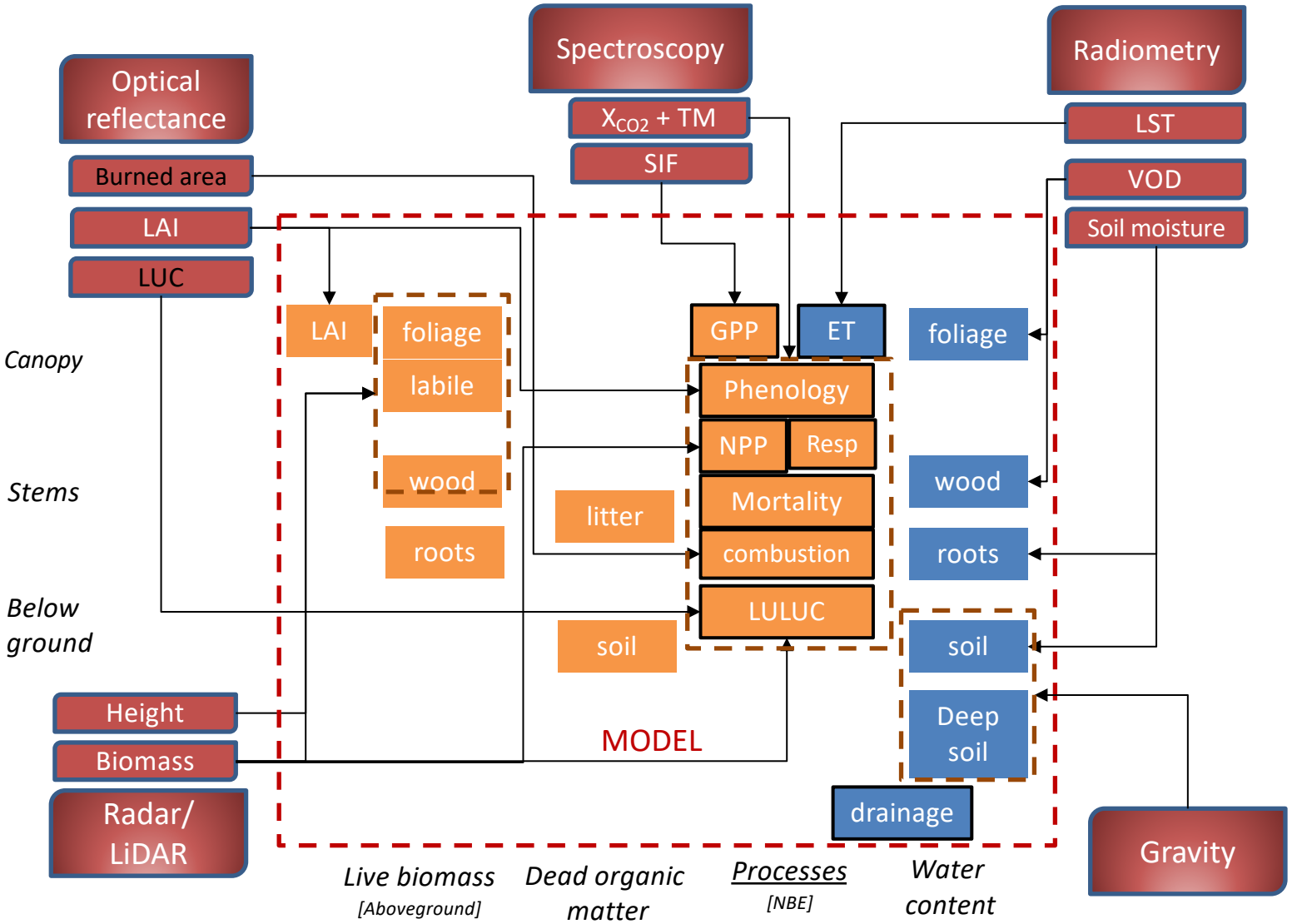
MDF via CARDAMOM to calibrate the model to reproduce biomass accumulation rate under historical climate and matching current LAI time series and soil C





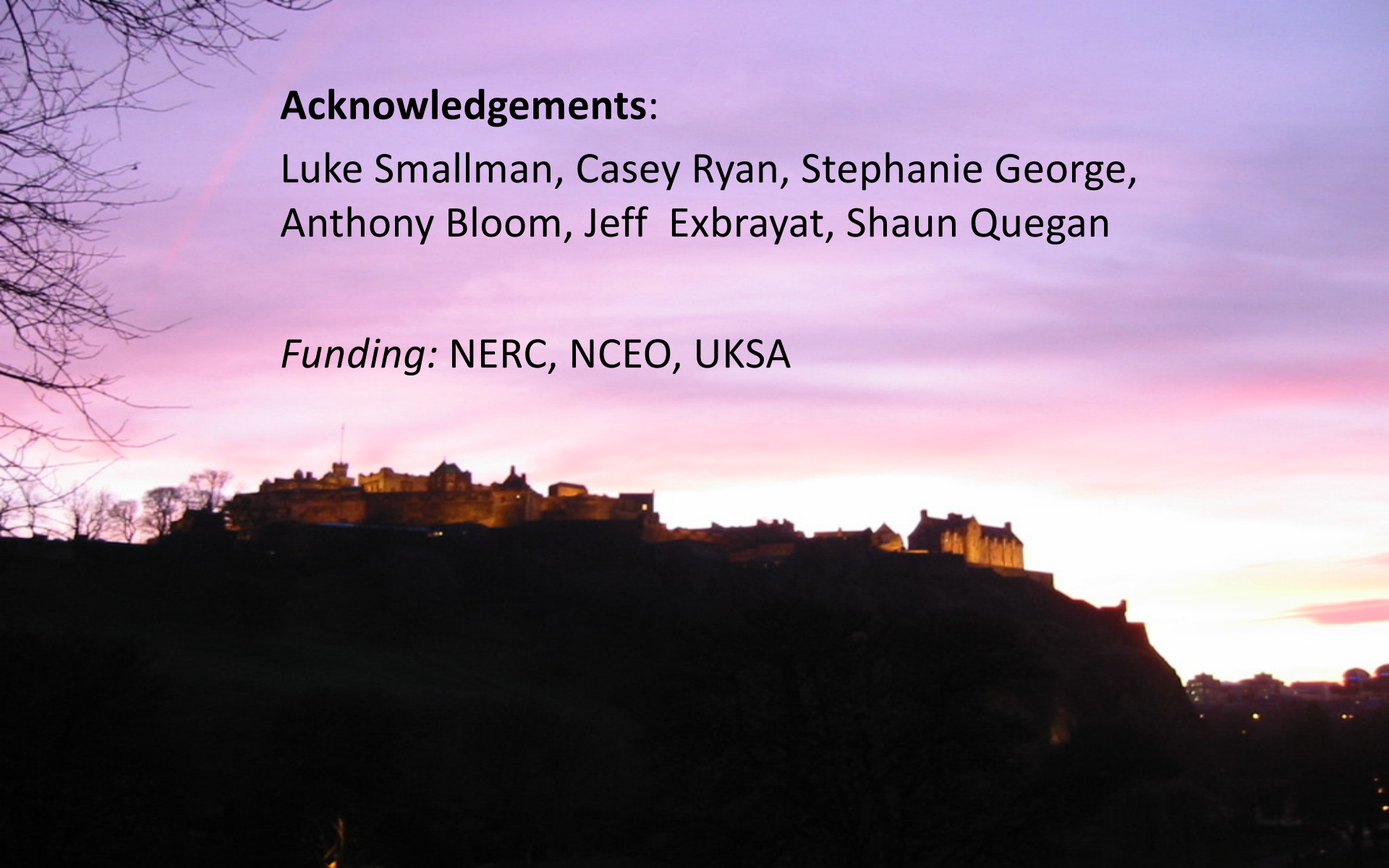


Model fit was only possible through introducing a density-dependent feedback on wood mortality


Optical reflectance
Burned area
LAI
LUC
Spectroscopy
X_{CO2} + TM
SIF
Radiometry
LST
VOD
Soil moisture
Canopy
Stems
Below ground
LAI
foliage
labile
wood
roots
litter
soil
GPP
ET
Phenology
NPP
Resp
Mortality
combustion
LULUC
foliage
wood
roots
soil
Deep soil
drainage
MODEL
Height
Biomass
Radar/ LiDAR
Gravity
Live biomass
[Aboveground]
Dead organic matter
Processes
[NBE]
Water content

Challenges

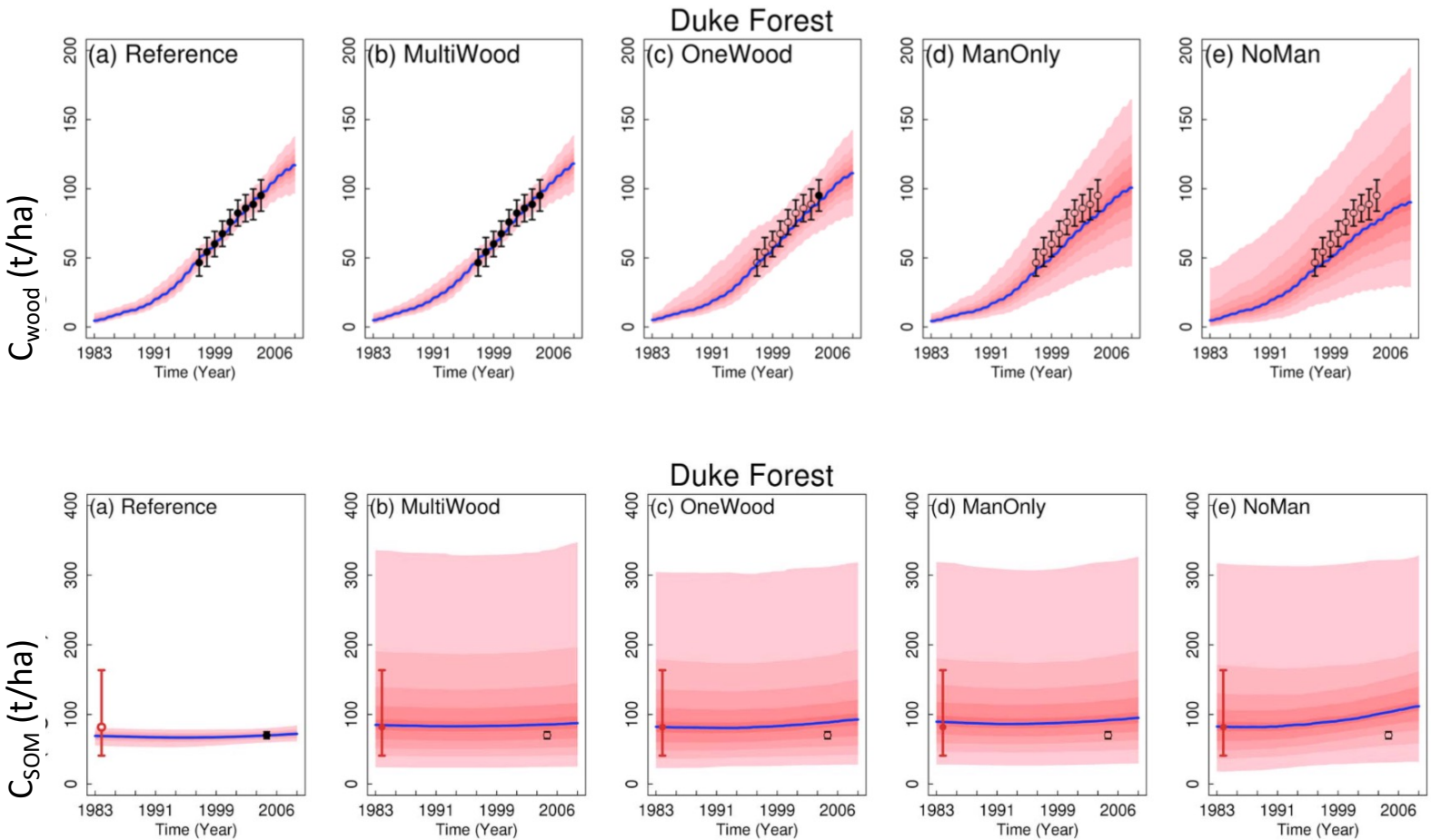
- What is the change detection accuracy?
  - Can we determine forest growth rates from space?
- Do we understand scaling error?
- Can error/bias be quantified accurately enough?
- Multiple missions -> Are products consistent? What is product longevity beyond each mission?



Acknowledgements:

Luke Smallman, Casey Ryan, Stephanie George, Anthony Bloom, Jeff Exbrayat, Shaun Quegan

Funding: NERC, NCEO, UKSA



Smallman et al (2017)

Using biomass distributions to assess MRT?

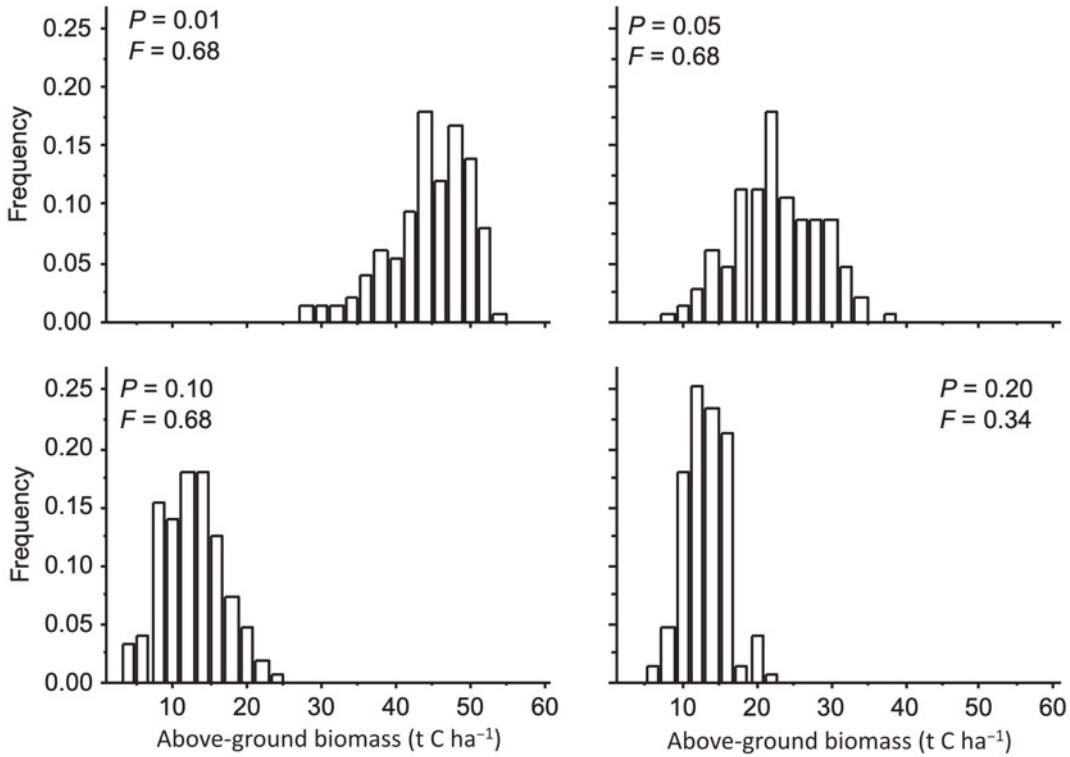


Figure 4. Simulated landscape variability in biomass stocks for a Mozambican calibration of A-DALEC under stochastic disturbance over 150 ha. The four panels show varying levels of disturbance probability (*P*) and intensity (F, fraction of C biomass lost in disturbance).

Williams et al. 2013

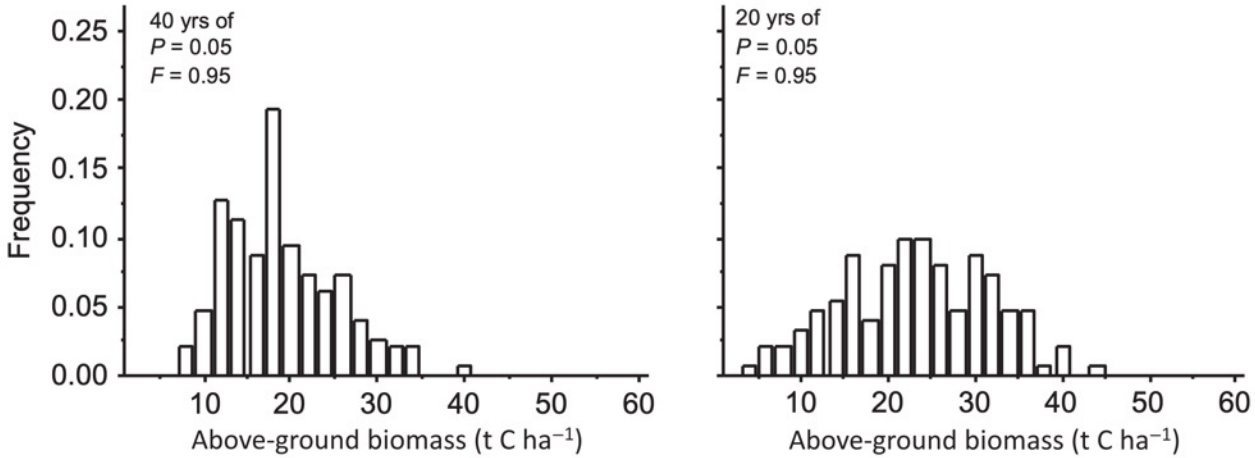


Figure 5. The effect of increased disturbance over 40 years (left) and 20 years (right) on a steady state forest structure resulting from lower disturbance ($P = 0.01$, $F = 0.68$, see top left panel in Figure 4). The histograms show the distribution of biomass after this period of increased disturbance.